



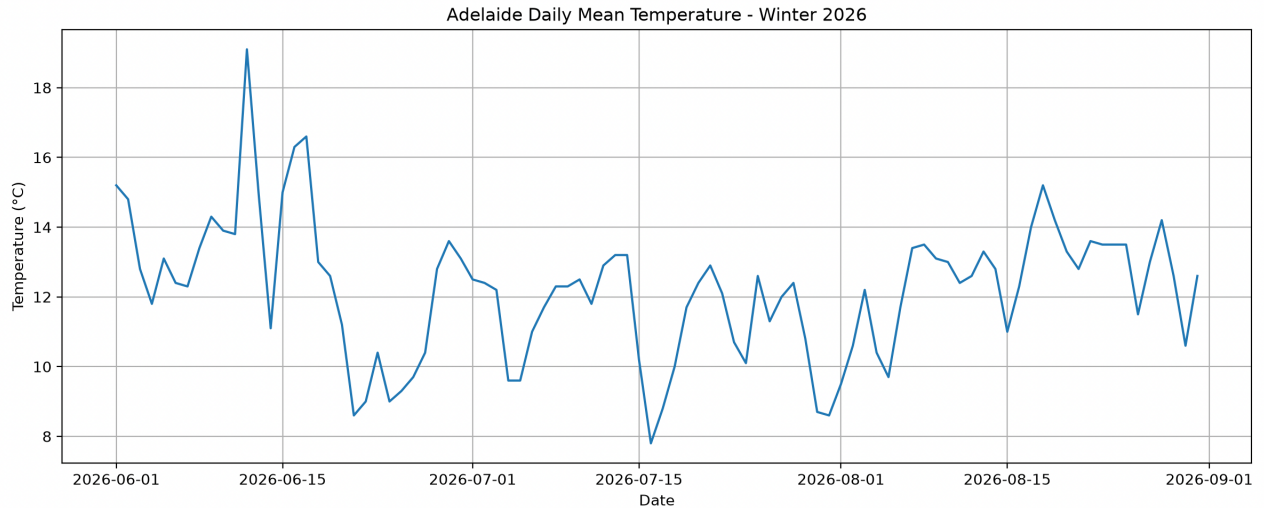
LAB 06 ANSWERS REPORT

HANS PUJALTE (PUJA0009)

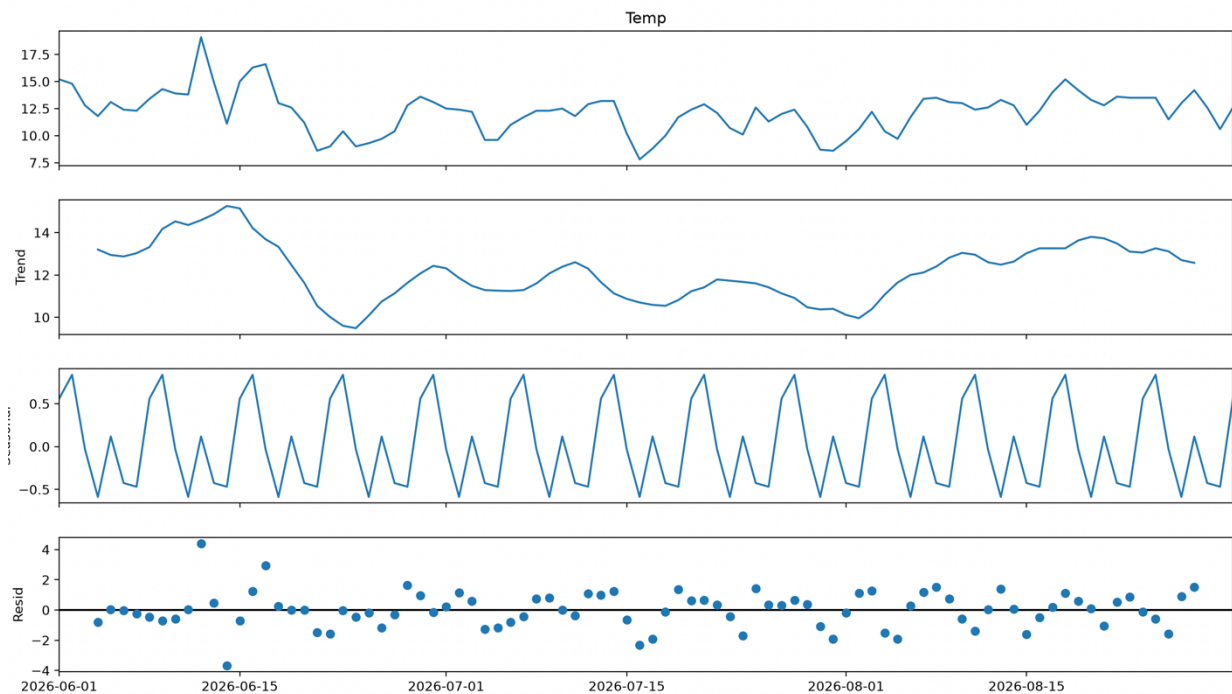
COMP3742, TONSLEY
COMPLETED: 18/9/2026

Hans Pujalte

1. Use the Open-Meteo API to fetch Adelaide's daily temperature data for Winter 2026 (June, July, August) and plot the time series. Identify any obvious trends or seasonality in the data. This data should be utilised for question 2-5 below.



Seasonal Decomposition of Adelaide Daily Temperature



Time series showcases short-term fluctuations in Adelaide's winter temperature, but a clear overall trend could not be identified across June to August'; temperature was relatively stable. However, there is a consistent seasonal pattern which is expected due to a recurring daily temperature cycle.

2. Fit an ARIMA model to the Adelaide daily temperature data for Winter 2026. Use the order (5,1,0) and calculate the training RMSE.

RMSE: 1.1215026010894689

3. Using the Adelaide daily temperature data for Winter 2026, compare the performance of ARIMA models with different orders: (5,1,0), (1,1,1), and (2,1,2). Calculate and compare the training RMSE for each model.

RMSE (5_1_0): 1.1215026010894689
RMSE (1_1_1): 1.1384821568491093
RMSE (2_1_2): 1.6265069038955522

Out of the three orders, the ARIMA (5,1,0) order model had the lowest RMSE value of 1.12. In comparison, (1,1,1) had a RMSE value of 1.14 and (2,1,2) with 1.62.

4. Perform a 7-step rolling forecast on a hold-out test set using the best ARIMA model from the previous question. Calculate the MAE, MSE, RMSE, and R^2 for the rolling forecast.

MAE: 0.937
MSE: 1.508
RMSE: 1.228
 R^2 : -0.251

5. Update the model to include an exogenous variable (e.g., hour of the day). Fit the model and compare its performance with the best Adelaide daily temperature ARIMA model without the exogenous variable

Ran out of time for this, Sorry!